

Computer Science 1016S - 2026 Notes to Students

Course Description

Computer Science 1015F/S and 1016S (or 1010H and 1011H) together constitute a complete Computer Science curriculum for first year students, offering an introduction to the development of algorithms and writing of computer programs, along with other selected topics in Computer Science. The course is offered in a blended learning format. Student contact time is in a single face-to-face lecture per week as well as a practical session aimed at reinforcing the principles introduced in the online lectures and giving students time to do exercises under the supervision of tutors.

Prerequisites

CSC1015F (or at least 70% for CSC1017F/S).

Staff

Course Convenor:	Mr. Aslam Safla	aslam@cs.uct.ac.za
Lecturers:	Prof. Sonia Berman Francois Meyer Asad Jeewa Unays Bhad	sonia@cs.uct.ac.za francois.meyer@uct.ac.za asad.jeewa@uct.ac.za BHDUNA001@myuct.ac.za
Teaching Assistants (TAs):	Sian Heesom-Green	hsmsia001@myuct.ac.za
Tutors:	(will be announced on Amathuba)	

Textbook and Notes

The highly recommended textbook is as follows:

Absolute Java, (International/Sixth edition or later) *by Walter Savitch*, Pearson / Addison-Wesley.

Class notes (copies of slides) may be available for selected sections and announced by the relevant lecturer(s). Electronic copies of lecture slides will be made available on Amathuba.

Amathuba

Amathuba (<http://amathuba.uct.ac.za>) is the university-wide online learning management system that gives you access to resources to assist in the learning process. The class website for all courses will be located on the Amathuba system.

Lecturers, TAs and tutors may be consulted through Amathuba – this is preferable since any questions that are answered may benefit other students as well. Amathuba is used for the submission of ALL practical assignments and practical tests and for providing students with marks for assignments and tests, and feedback where appropriate. All students will be expected to consult the website on a daily (Monday-Friday) basis for updates on assignments, marks, hints, deadlines, etc.

Please refrain from posting anything of the following nature anywhere on the website, as it may violate the university's Appropriate Use of Computer Facilities policy (see ICTS website), necessitating disciplinary and/or legal proceedings: sexist, racist or otherwise discriminatory

comments, flame wars, troll; segments of program code (other than something provided by the instructor) ; solutions to graded work (or part thereof) before or after submission; or illegal material.

Lectures

Lectures are held in the following venues:

	Venue
Period 4:	JD LT2
Period 5:	JD LT2

Lectures are held from Monday-Wednesday every week. You should attend **one** of these lecture sessions per week (see Groups on Amathuba to signup for your session).

Continuity of Teaching

If lectures are disrupted and / or access to campus is restricted, teaching and learning will continue online, with live online lectures via MS Teams. All lectures will be recorded as well.

The course convener will make announcements about changes to the mode or timing of any activities associated with the course (lectures, practicals, tests, ect.). Students without Internet access at home should take note of eduroam locations (<https://eduroam.ac.za/>) and other places where free wifi is available.

Tests

There will be 2 closed-book 45 minute theory tests. All theory tests will be conducted over Amathuba. Venue details will be communicated to you closer to the test.

There will be 2 open-book 40 minute practical tests. Each practical test will be offered 2 times – you may write one or both of them and the maximum mark obtained *for each first try in that test week* will be used for each test.

See the schedule for test dates.

Assignments

Assignments will be done on your own laptop computer. All Assignments will be submitted to the Automarker on Amathuba, except the Final Assignment (SIPP Essay), which will be submitted to the Assignments tab on Amathuba.

Questions and Submission

All questions for assignments, along with all related files, will be available on the class website on Amathuba. Practical assignments must be submitted electronically via Amathuba ONLY. The online submission system used to receive your assignments will provide the official timestamp used to determine whether a program is on time. Marks will be deducted automatically for automatically-marked assignments that are submitted late.

Marking

Most assignments will be marked automatically based on test cases and the marks will be uploaded to Amathuba. Tutors will manually mark randomly-chosen practical assignments during the semester.

Equipment and Programming Language

All programming will be done in Java (JDK v1.8.0 or later) unless otherwise stated. The tutors will compile and test programs on JGrasp running on MS-Windows. Assignments that are automatically marked are marked using JDK v1.8.

It is the responsibility of the student to submit a program that will successfully compile and execute on the specified platform. Any student who works on their own equipment must ensure that all assignments will compile and execute on the university equipment before submission – no discussion will be entered into after submission.

It is ALWAYS the student's responsibility to ensure that adequate backup copies are made of all work in progress and all work already completed. Loss of data or programs is not an acceptable excuse for non-submission or late submission of assignments.

Practical Sessions

Practical sessions are held either in one of the SCILABs or in a lecture room. You must attend one 2-hour session each week, where you may work on the current practical assignment and discuss any general issues related to practical work with the tutors who are available. **Students can complete a weekly compulsory practical assignment during these sessions, and students write a practical test (as described above) if one is scheduled for that week.**

It is your responsibility to ensure that you choose a practical session (on Amathuba) that fits in with your timetable. If you discover that all slots that suit you best are full already, it is your responsibility to find a student with whom you can arrange a swap as soon as possible.

Plagiarism and Academic Dishonesty

Refer to attached document on the departmental plagiarism policy, especially regarding academic dishonesty and computer program submissions. This policy will be strictly enforced.

All assignments, tests and examinations are done individually – there is NO group work allowed in this course. It is acceptable to discuss the questions for assignments with peers but not the specific details of the solutions. When in doubt, speak to a tutor or TA.

DP Requirement

A student is granted DP status (and may write the exam) in CSC1016S if the following conditions are met:

- subminimum mark of 50% for practical test average
- $(3/5 * \text{Practical average} + 2/5 * \text{Practical test average}) \geq 45\%$,

Final Examination

The examination timetable will be published sufficiently in advance of the final examination. It is the student's responsibility to take note of the correct time and place for the examinations.

The examination will be cumulative, closed-book and closed-notes (i.e., you may not bring your notes or textbooks into the examination room), and 2 hours in duration.

A final mark in CSC1016S will be calculated as follows:

$$\text{Final} = 0.10 * \text{Practical average} + 0.15 * \text{Test average} + 0.15 * \text{Practical test average} + 0.60 * \text{Exam}$$

In order to pass, ALL of the following requirements MUST be met:

- Final $\geq 50\%$
- subminimum mark of 45% for practical test average

- $(3/5 * \text{Practical average} + 2/5 * \text{Practical test average}) \geq 45\%$
- $(1/5 * \text{Test average} + 4/5 * \text{Exam}) \geq 45\%$

Refer to the CSC1011H notes to students for the CSC1011H final mark calculation.

Supplementary Examinations

Students who do not pass but obtain a mark of 45-49 may be awarded a supplementary examination; this is indicated either with xS (with $45 \leq x \leq 49$) or OSS ['failed theory subminimum'] in PeopleSoft. These are written in January and the exams office will contact those eligible for it, including the exact time and date of the supplementary exam.

The final mark will be calculated as above, but then with the supplementary exam mark. The system then converts this into a “UP” unqualified pass or “UF” unqualified fail if you had a plain supp and into the 'June grade' if you had an OSS with the June exam.

Please also note the following (short version of some of the handbook rules): there is no supp on a supp exam, no deferring a supp exam, no supp on a deferred exam, and no deferring a deferred exam.

Grade Allocation

1 = 75-100; 2+ = 70-74; 2- = 60-69; 3 = 50-59; F = 0-49

Information Dissemination and Communication

Attendance and Absence

This is a face-to-face lecture course. While attending the lectures is not mandatory after the first day, all marked work (assignments, tests and quizzes) will be based on the lectures. Obviously, non-attendance at tests will result in a mark of 0 (zero).

Medical Notes and Short Leave

All students are expected to do all assigned work. If a student falls ill, they must submit a medical certificate to the course convenor as soon as possible.

A valid medical certificate must be signed by a **qualified medical doctor** and clearly state the days on which you were medically unfit to work. Please submit a copy of the original medical certificate via email to sick-csc1016s@cs.uct.ac.za .

In other/exceptional circumstances, where a student would like to be away from their studies for a short and defined period of time, the Short Leave Application form (under Amathuba resources) must be filled out and submitted to the course convenor for recommendation and HoD for approval.

Appeals Procedure

If a student feels that their marks are incorrect for any piece of marked work, they must first approach the tutor or TA responsible for that test or assignment. Then, if need be, they may appeal to the course convenor. Finally, they may appeal to the HoD.

Queries

Any queries about the content of the lectures **MUST** be directed to the lecturer teaching that section.

Any queries about marks or marking of practical assignments must be directed to your tutor or TA.

All marked work (whether in paper or electronic format) must be kept until the end of the semester. In general, queries about marks **MUST** be made within a week of marked work being returned. No queries about any marks will be entertained after the final examination.

Any queries about the administration of the course must be directed to the TA.

The course convenor must **ONLY** be contacted as a last resort unless otherwise indicated.

Syllabus

Corresponding chapter numbers in the prescribed textbook (4th edition) are indicated in parentheses.

- Introduction to Object Oriented Programming (OOP), Java syntax (Chapter 1/2/3)
- OO Design, UML I, Classes and Instances (Chapter 4/12)
- Methods, Variables, Constructors (Chapter 4)
- Statics, References, Aggregation/Composition (Chapter 5)
- Inheritance, Polymorphism, UML II (Chapter 7/8)
- Testing with Junit
- Abstract data types, Interfaces, Generics, Design Patterns (Chapter 12/13/14)
- Arrays, Linked lists (Chapter 6/15.1/15.2/15.3)
- Stacks, Queues (Chapter 15.4)
- Graphics, GUIs (Chapter 17/18)
- Human Computer Interaction

Schedule of Lectures and Practical Work

Week	Date	Monday	Tuesday	Wednesday	Thursday	F2F Lecture	Prac. Test	Assignment (subject to change)
1	27-Jul	Intro & Java Basics	Java Basics	Java Basics	Java Basics	Intro		
2	03-Aug	Java Basics	Java Basics	OOP Principles	OOP Principles	Java1		1 - Orientation & Basic Syntax
3	10-Aug	Holiday	OOP Principles	OOP Principles	OOP Principles	Java1		2 - Creating a class with methods
4	17-Aug	OOP Principles	HCI	HCI	HCI	Java2		3 - Creating a complete class (Constructors + Junit, etc.)
5	24-Aug	HCI	Static	References (Theory_Test 1 (26 Aug, 6pm))	References	HCI	PT1	4 - Encapsulation - Get/Set
6	31-Aug	ARRAYS	Inheritance	Inheritance	Inheritance	Java2	PT1	5 - References & Consolidation
	07-Sept							
7	14-Sept	Polymorphism	Polymorphism	Polymorphism	UML	Java2		6 - Arrays & Inheritance
8	21-Sept	UML	UML	UML	Holiday	Java2		7 - Inheritance, Polymorphism & UML
9	28-Sept	Abstract	Interface	Interface	Generic	UML		8 - D-School?
10	05-Oct	Inner Classes	Linked List	Linked List	Stack & Queues	Java3		9 - Abstract, Interface, etc.
11	12-Oct	GUI	GUI	GUI + Theory_Test (14 Oct, 6pm)	GUI	Java3	PT2	10 - GUI
12	19-Oct	GUI	GUI	UML	UML	GUI	PT2	11 -Linked List
13	26-Oct	HCI	HCI	Consolidation				